

A 3D CAD rendering of a complex scientific instrument, the Hall A Moller Polarimeter 12 GeV Upgrade. The device consists of a long, horizontal assembly of various components, including a large central rectangular block, several smaller cylindrical and conical parts, and a tall vertical structure on the right side. The entire assembly is supported by a robust metal frame with multiple vertical and diagonal beams. The rendering is in a light blue/gray color scheme.

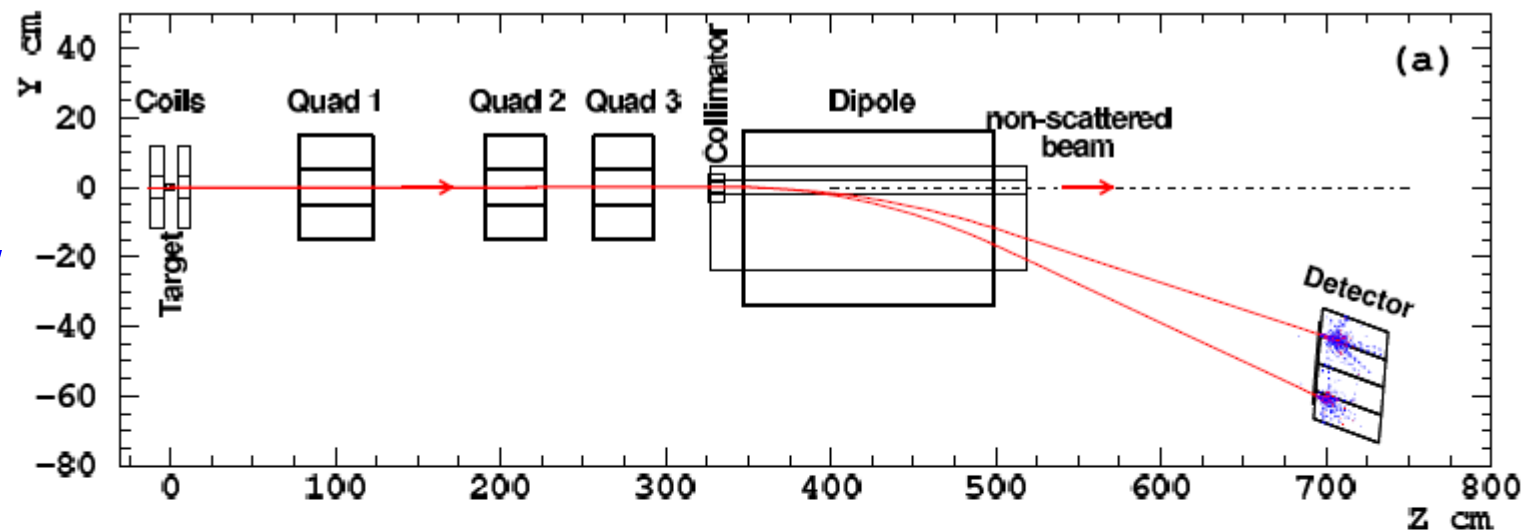
Hall A Moller Polarimeter 12 GeV Upgrade

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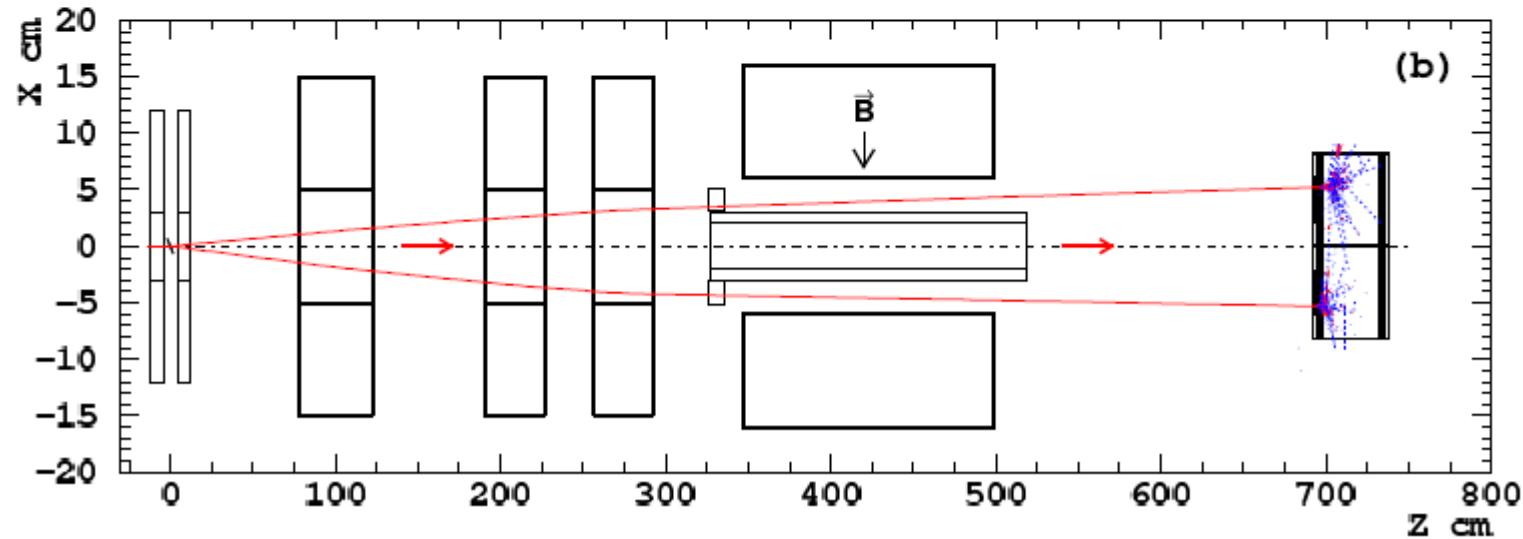
Hall A Collaboration Meeting, Dec 2012

Hall A Moller Polarimeter for 6 GeV

Side View



Top View



$$P_{\text{beam}} = \text{Asym} \times P_{\text{target}} / \text{AnaPower}$$

Designed to detect electrons of 0.8-6 GeV, $E' = E_{\text{beam}}/2$

The Low Field Target

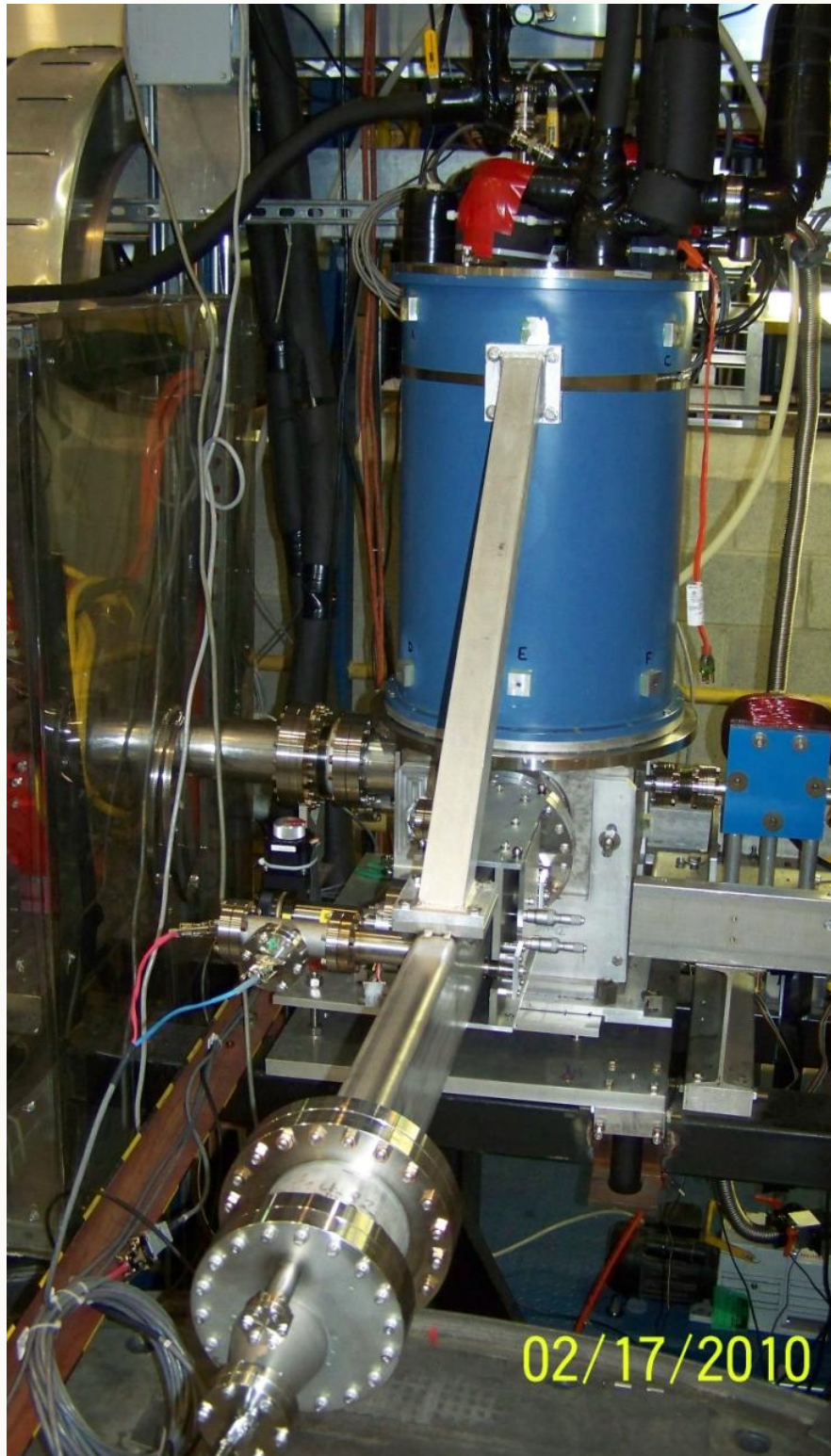


- Sys error $\sim 1.7\%$
- 5 target foils of $15 \times 3 \text{ cm}^2$
- Supermendur and iron foils, $7\text{-}30 \mu\text{m}$ thick
- 20° tilted



Foil	0	1	2	3	4	5
Material	Al	SM	SM	Fe	Fe	SM
Thickness μm	16.5	13.0	29.4	14.3	9.3	6.8
$\mathcal{P}_{center}$ % prelim	-	7.95	8.20	7.62	7.42	7.97

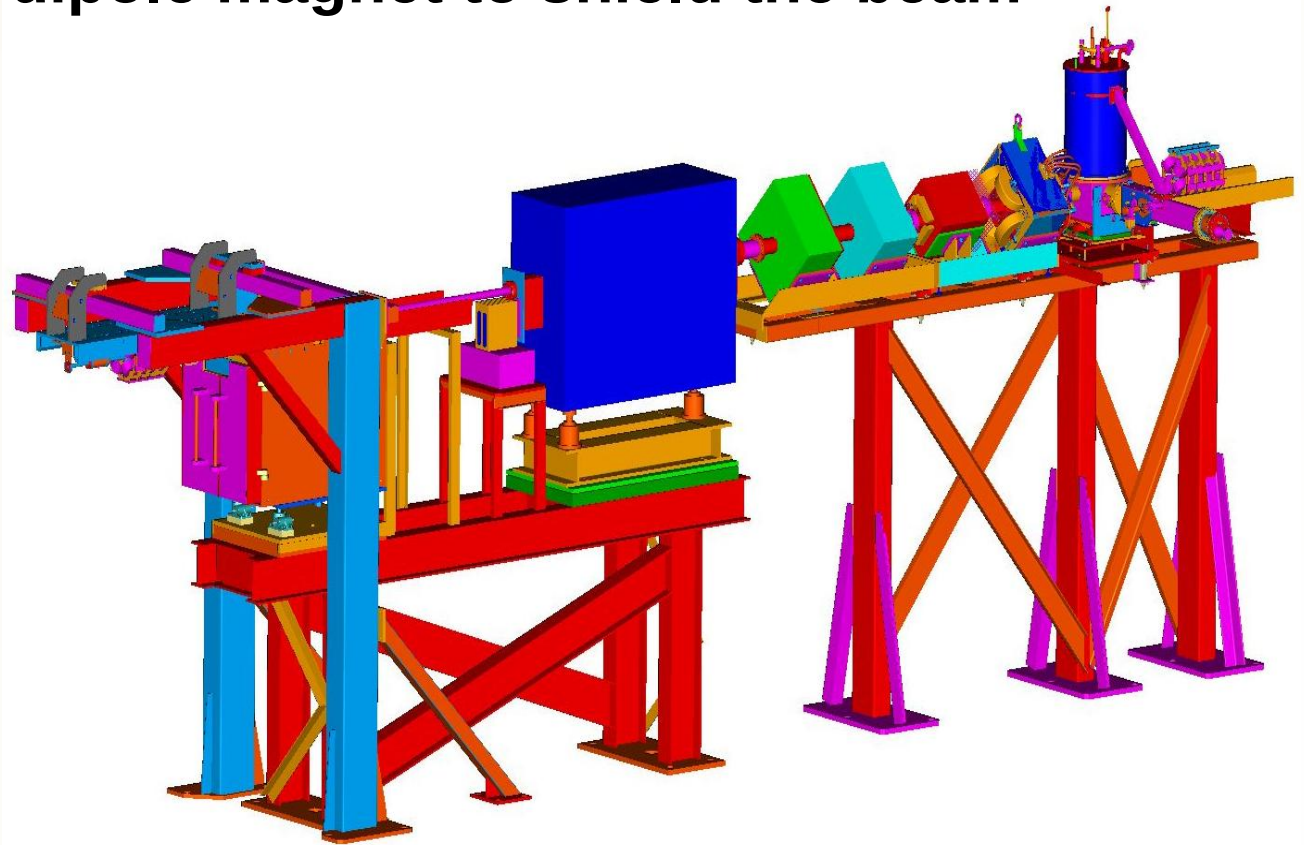
The High Field Target



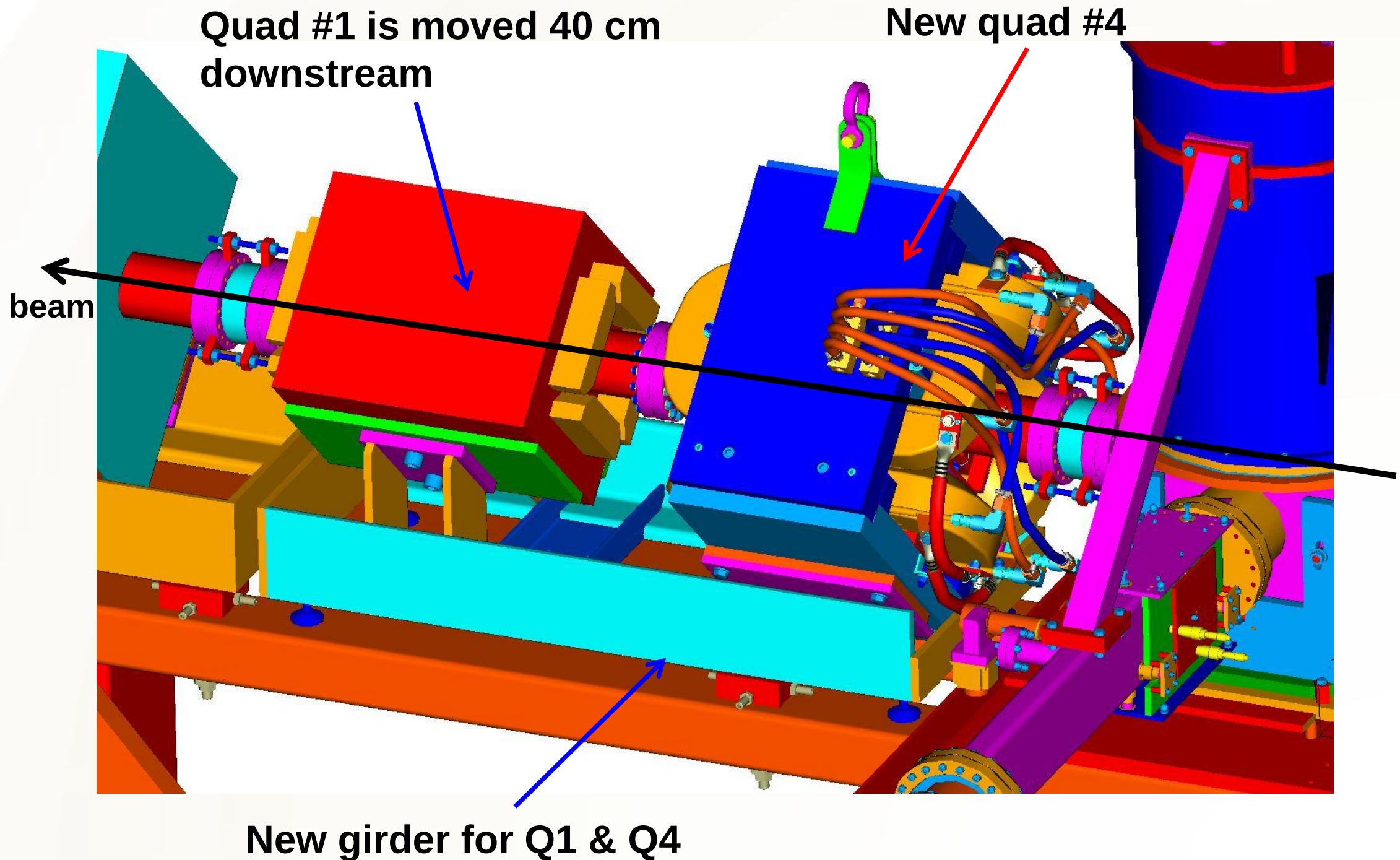
- Sys error $\sim 0.9\%$
- 4 Tesla maximum field
- To use this target, we should:
 - a. Improve the target holder for better alignment
 - b. Build Cryogenic line to feed the magnet
 - c. Change target foils

Upgrade for 12 GeV

- Add the 4th quadrupole magnet to cover electron energy range 0.8~11GeV. Need to shift Q1 and a new bench for Q1 and Q4
- Add a shielding pipe into the dipole magnet to shield the beam from high dipole field
- Møller electrons bending angle changes from 10° to 7.3° in order to use the same dipole power supply
- Rearrange the beam line shielding downstream of the dipole for 7.3° angle
- Lift up the detector shielding box by 10cm to match 7.3°
- Need a new detector shielding house (can not cut from the old one)
- New correctors and new BPM downstream of the Møller dipole



Changes to the Quadrupole Magnets



Status of the New Quadrupole Magnet (Q4)



- The new quadrupole magnet (Q4) has already been designed and manufactured
- New bench for Q1 and Q4 has also been designed and manufactured
- The magnetic field of Q4 was mapped by magnetic group in the summer of 2012
- Q4 and the bench has already been installed in the beam line

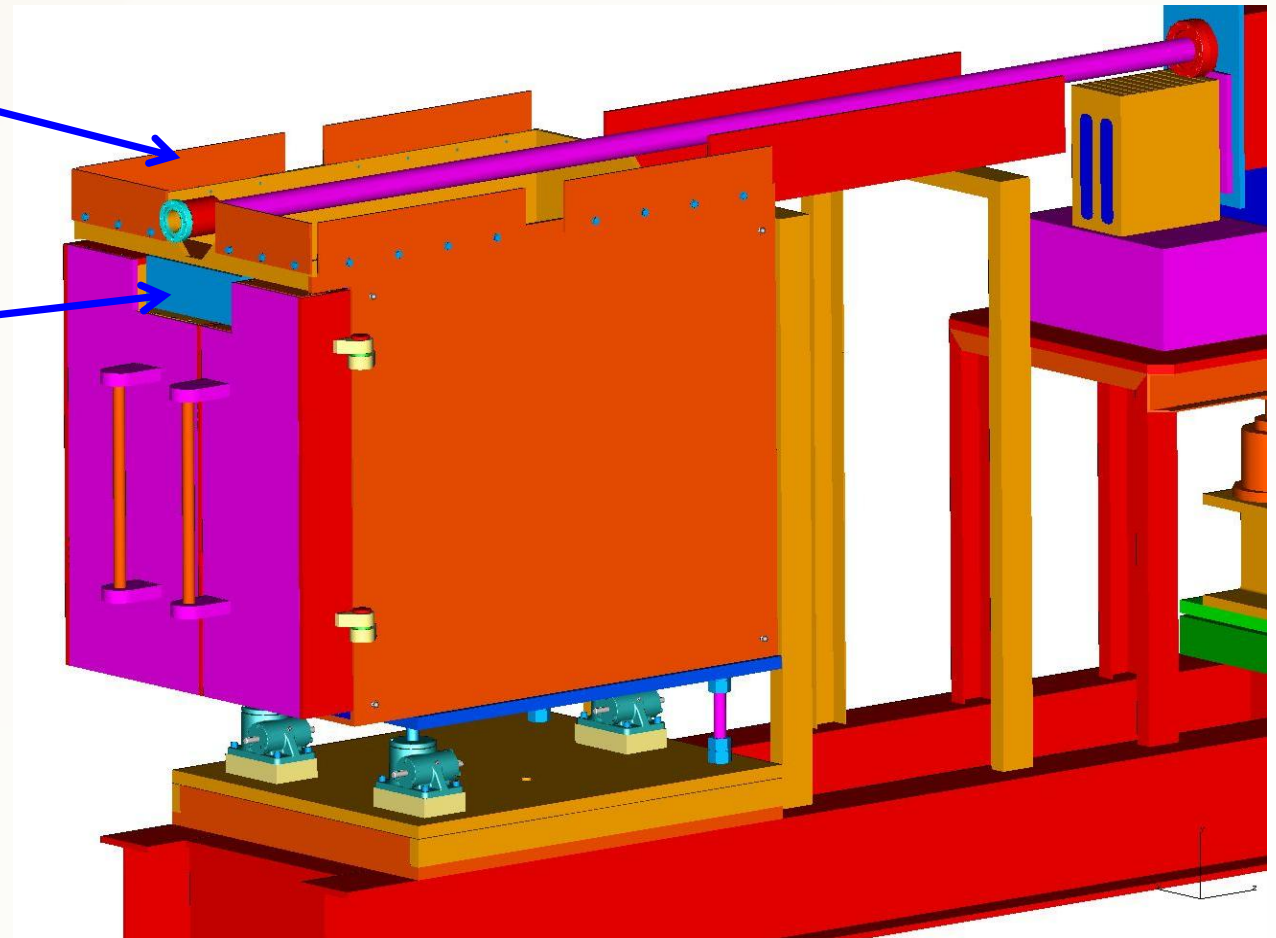
Changes to the Dipole Magnet



- Beam line Shielding pipe inserted into dipole magnet to increase to the field attenuation.
- Rearrange for 7.3 degrees
- Overall length of the pipe is 212.4 cm.
- Maximum beam shift in vertical is less than 2 mm at the pivot or 9 mm at the beam dump with 11 GeV beam .

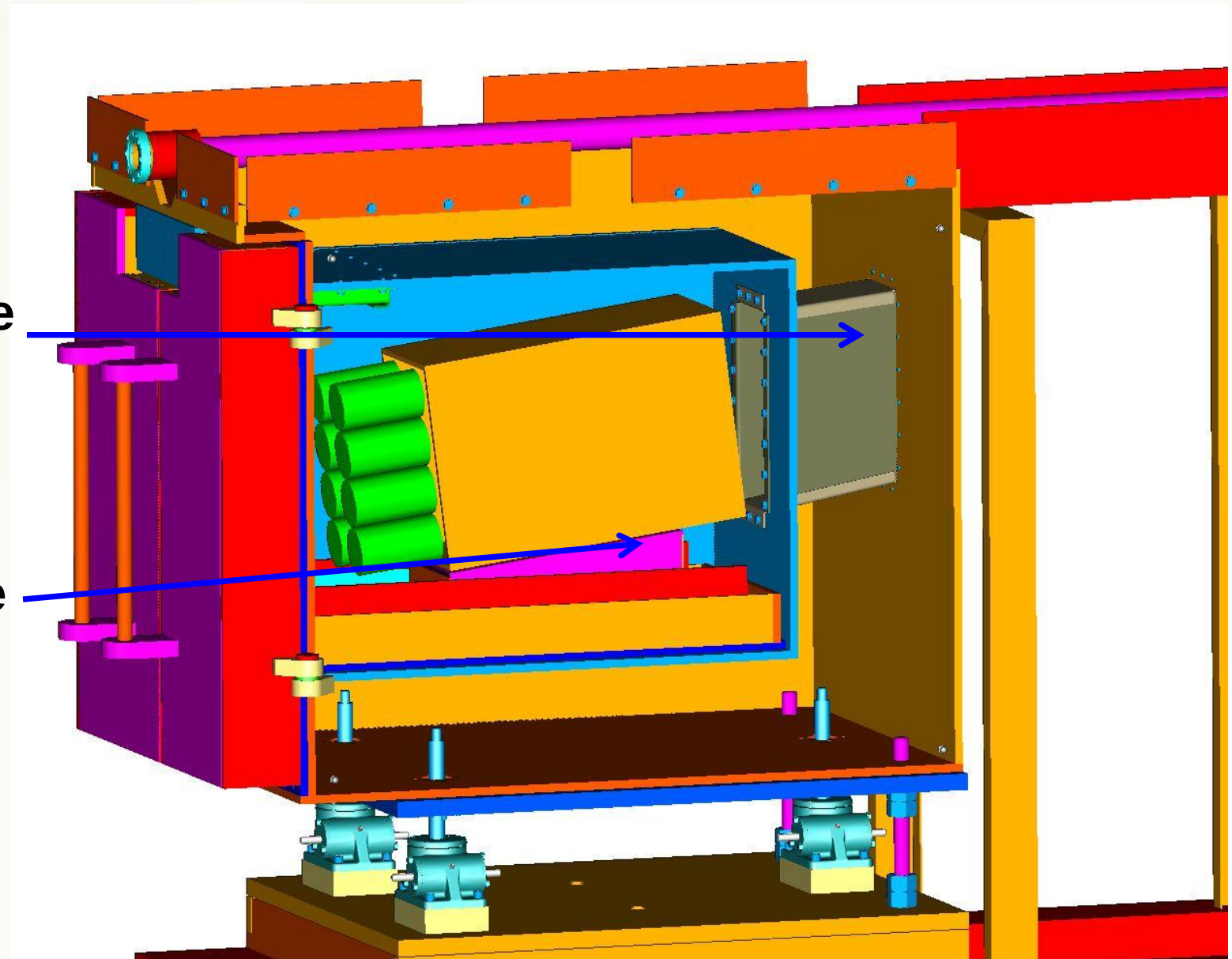
Changes to the Detector Shielding Box

- Top of outer box extended to safely contain lead shielding bricks.
- Clearance in doors for upside down girder components.
- Detector shielding box lifted 7 cm above 6 GeV position. (Designed to be lifted by 10 cm if needed.)



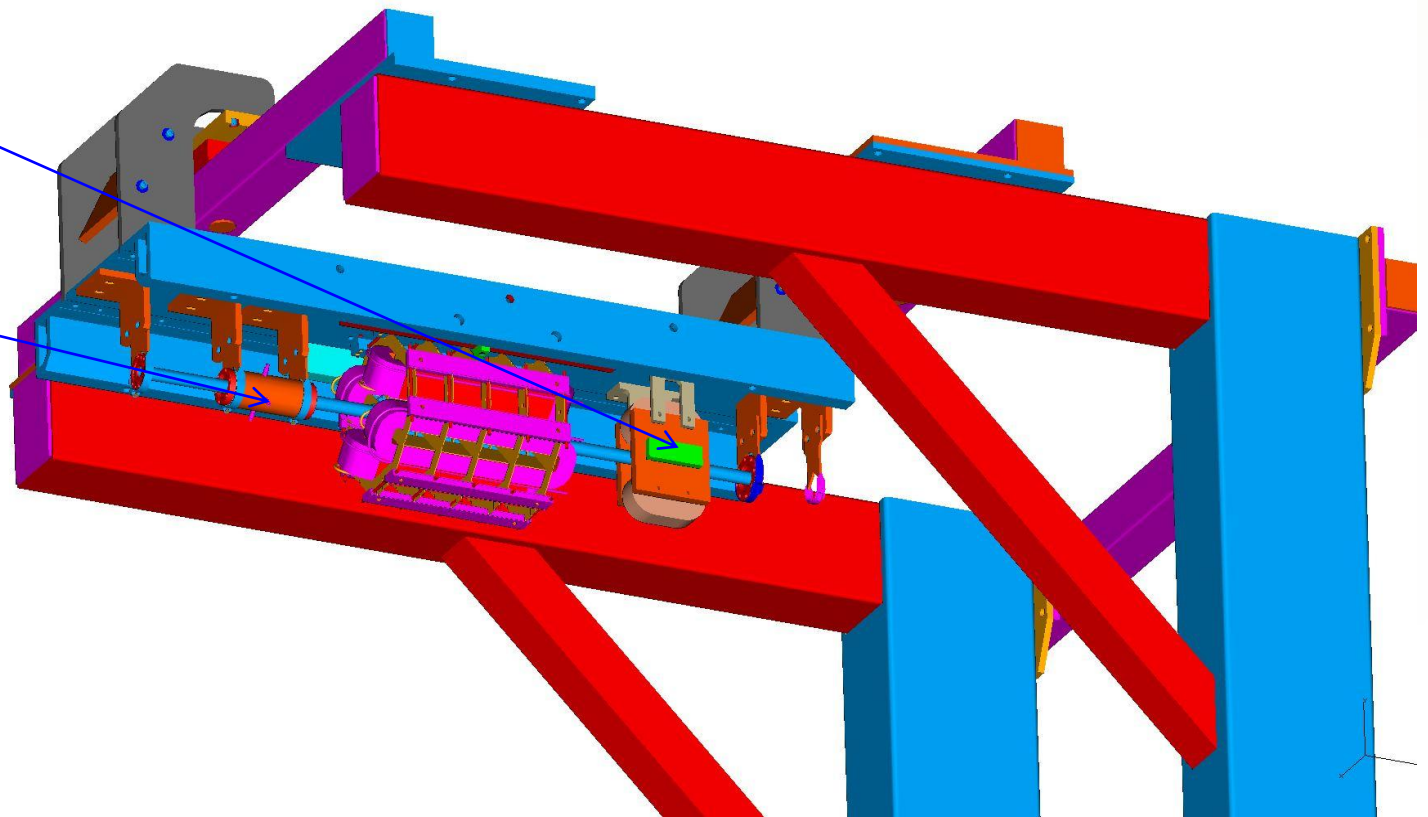
Internal Changes to the Detector Shielding Box

- Added a sheet metal enclosure around the opening of the box.
- Changed the mounting angle of the detector package from 10 deg to 7.3 deg.

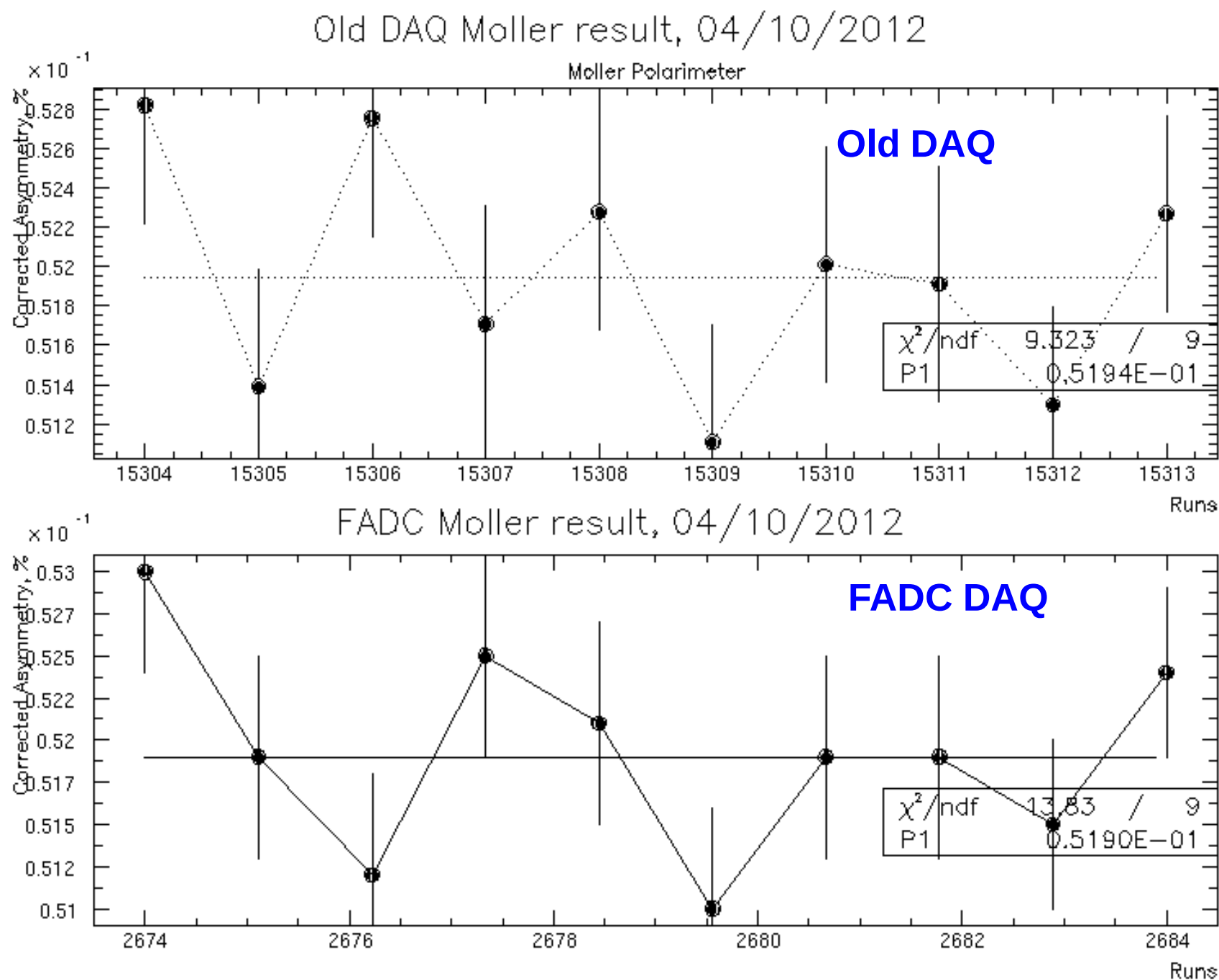


Changes to the Upside Down Girder

- Removed one quad
- Move the remaining quad to a central location
- Added vertical corrector magnet
- Added a BPM



Comparison between old and FADC DAQs



Discrepancy is about 0.1-0.5%, using scaler triggers.

Systematic Errors for Current System

Variable	"Low" field	"High" field 3.9T
<i>Target polarization</i>	1.5%	0.35%
<i>Analyzing power</i>	0.3%	0.3%
<i>Levchuk effect</i>	0.2%	0.3%
<i>Target temperature</i>	0.02%	0.02%
<i>Dead time</i>	0.3%	0.3%
<i>Background</i>	0.3%	0.3%
<i>High beam current</i>	0.2%	0.2%
<i>Others</i>	0.5%	0.5%
<i>Total</i>	1.7%	0.9%

Electronics Upgrade Goals

1) Increase detector bandwidth up to 200MHz

- Programmable Logical Unit(PLU) module LeCroy-2365, bandwidth <75 MHz, CAMAC (**out of market now**)
 - ← replaced by PLU module based on CAEN V1495 board (bandwidth 200MHz, VME, FPGA)
- Discriminator Ortec-TD8000, input rate <150 MHz, CAMAC
 - ← replaced by P/S 706 (300 MHz, NIM), modified for remote threshold setup with DAC type of VMIC4140

2) Reduce readout time from ADC and TDC modules

- ADC LeCroy 2249A, 12 channels, CAMAC
 - ← replaced by QDC CAEN V792 (32 channels, VME)
- TDC LeCroy 2229, CAMAC
 - ← replaced by TDC V1190B (64 channels, 0.1ns, VME)

3) We also consider to implement some new triggers unit (for ADC, TDC trigger and gate).

Electronics Upgrades Status

1. Some electronic modules have been installed to VME and NIM crates for testing.
2. Firmware for PLU module is under developing and testing.
3. Software:
 - Library to control DAC for remote threshold setup is ready.
 - Library for configuring and controlling of PLU module is ready (Logical Unit and Scalers).
 - Readout list for TDC and PLU Scalers is ready.
 - Coda for testing TDC and PLU modules is running.

ToDo List.

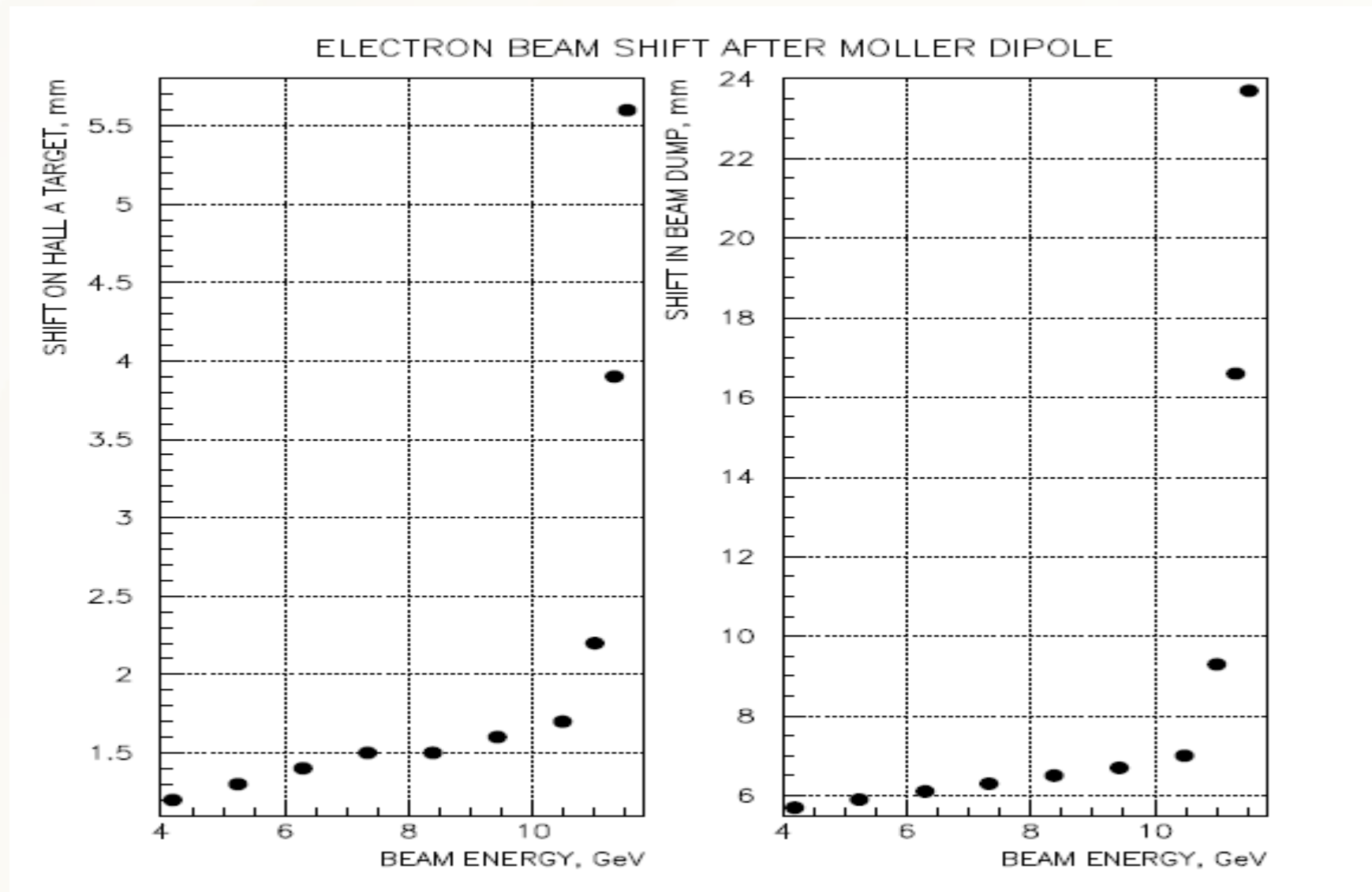
- Code to analyzing Data Trigger of FADC DAQ
- PLU Firmware optimization
- Continue testing of PLU+TDC+QDC with CODA and Analyzer (modify from old DAQ).
- Migrate both old DAQ and FADC DAQ to CODA 3 (configuration files, mSQL DB, CRL code)

Summary

- **Current Moller polarimeter working very well. Discrepancy between old and FADC DAQ is 0.1-0.5%.**
- **The systematic uncertainty of moller measurement with low field target is ~1.7% and 0.9% for high field target.**
- **Engineeering designs has been finished. New Q4 has been mapped. Q4, Q1 and their new bench have been installed.**
- **Changes to the Dipole, the detector house and the upside down girder have been done in the hall.**
- **Electronics module test is in progress.**
- **Software upgrade is going well.**

Back Up

Vertical Beam Position Shifted Due to the Moller Dipole Magnet



TOSCA calculations for the electron beam shift after the Møller dipole with shielding insertion and additional shielding pipe